Response to Reviewers: CAISAIS 2026 Revision

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General Remarks

Thank you to both reviewers. Changes in the manuscript are marked in blue.

Reviewer 1

Comment 1.1

Does the driver monitoring evaluate speeding? If yes, up to what speed can the detector be considered accurate?

Response. No. The detectors do not evaluate speeding or estimate vehicle speed. They localize visible driver cues in single frames (four in RGB, two in NIR), and no result is stratified by driving speed, so there is no speed range to report. (If inference speed was meant, FPS and p50 latency for RGB are in Table III, and p50 latency for NIR is in Table IV.)

Comment 1.2

What do the authors mean by 'negative'?

Response. A negative is a frame (RGB) or snippet (NIR) with no target cue, i.e., zero ground-truth boxes: none of the four cues in RGB, and no drinking or phone-use tracklet in NIR. Any above-threshold detection on one is a false detection, which is what FD_{100} counts. The definition is now in the Abstract, the RGB protocol, and the NIR protocol.

Comment 1.3

Please describe the potential benefits that this study may offer to the community.

Response. Added at the end of the Introduction: a reusable, reproducible subject-disjoint protocol with publicly archived annotations, partitions, and checksums; deployment-oriented evidence on accuracy, false-detection, and latency trade-offs among lightweight detectors; evidence that pooled metrics can hide difficult drivers; and exploratory evidence that more negative training examples do not consistently help.

Reviewer 2

Comment 2.1

The main limitation is that conclusions rest on a single fixed split with only three test drivers, and three of the four cues have very few test instances (28–56), so the differences between models are small relative to their spread. For example, D-FINE-N's macro-F1 lead (84.33 ± 4.51 vs 83.42 ± 1.36) is not distinguishable, . . .

Response. Agreed. The D-FINE-N macro-F_1 lead over YOLO11n is 0.91 points against seed SDs of 4.51 and 1.36, so it is now described as within seed variation in the Abstract and as not distinguishable in the Results. The micro-F_1 margin (4.54 points) is larger than the SDs but still rests on three seeds and three test drivers. Interpretation Scope now says that, with one split and 28–56 test instances for three cues, small differences are descriptive trade-offs, not rankings. Other subject partitions mean retraining every system, so they remain future work.

Comment 2.2

. . . and the abstract's claim that it has the best 'calibrated-threshold detection performance' should mention that it also has the worst FD100 (0.87).

Response. Agreed. D-FINE-N has the highest (worst) FD_{100}: 0.87 ± 0.21 vs. 0.63 ± 0.04 for YOLO11n and 0.67 ± 0.06 for YOLO26n. The Abstract now says so, and so do the Results and the Conclusion.

Comment 2.3

In the NIR protocol, the statement that 'validation and test contain identical ordered examples' needs clarification, because if the two sets are the same, the thresholds were effectively chosen on the test set, which contradicts the protected-test claim.

Response. The wording was unclear, and the two sets are not the same. The 9:3:3 split puts 9, 3, and 3 of the 15 drivers in training, validation, and test, so validation and test are disjoint groups of three drivers, and thresholds were selected on validation data only; the protected-test claim therefore holds. The sentence meant that all sixteen runs use the same examples in the same order. Reworded in the NIR protocol.

Comment 2.4

The NIR study also changes the number of negatives and the number of training updates at once and uses one seed, so it can only show that the effect depends on the model, as the authors acknowledge.

Response. Agreed. The NIR protocol already notes that 1:6 has three times the unique negatives and about 2.31 times the optimization updates, and presents the study as exploratory training-negative exposure, not a class-ratio ablation. To make the limits explicit, the Abstract now says "single-seed" and "indicates" (previously "confirms"), and the Conclusion calls it an exploratory single-seed experiment in which negatives and training updates vary together. No new NIR runs.